

Data Acquisition and Processing of the GeoDAWN Aeromagnetic and Radiometric Survey

Task Order No. 140G0221F0024

NV_WestCentral_Geophysical_2020_D21



Contract No. G16PC00016

U. S. GEOLOGICAL SURVEY

NATIONAL GEOSPATIAL TECHNICAL OPERATIONS CENTER

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Introduction

Task Order Task Order No. 140G0221F0024 NV_WestCentral_Geophysical_2020_D21 was issued in March of 2021 to NV5 Geospatial, Inc. and its sub-contractor EDCON-PRJ, Inc. The GeoDAWN Survey is one of many geophysical surveys recently funded through the Earth Mapping Resources Initiative (Earth MRI). The contract covered the acquisition and processing of aeromagnetic and radiometric data acquired over a portion of west-central Nevada (**Figure 1**).



Figure 1. Location Map of the GeoDAWN Project Area.

The GeoDAWN survey area encompasses 51,857 square kilometers in west-central Nevada. It includes portions of Churchill, Esmeralda, Eureka, Lander, Lyon, Mineral, Nye, Pershing, and Washoe Counties, as well as small portions of Mono and Inyo Counties in California. The survey extends 330 kilometers from west to east and 370 kilometers from north to south.

A total of 149,030 linear kilometers of high-resolution aeromagnetic and radiometric data was acquired for the purpose of imaging the subsurface geology and structure of the area and to aid in identifying potential locations of undiscovered mineral and geothermal resources. These data will be used to evaluate the potential for undiscovered resources and will contribute to USGS and DOE-related mineral and geothermal resource assessments of the region.

Stakeholders for this project include the USGS Geology, Minerals, Energy, and Geophysics Science Center, the DOE Office of Energy Efficiency and Renewable Energy, the Nevada Bureau of Mines and Geology, Tribal Councils, and other state and federal offices.

The project was managed by the designated USGS Point-of-Contact (POC) from the National Geospatial Technical Operations Center in Rolla, Missouri. Chase Thompson was the Project Manager for the NV5 Geospatial GPSC3 contract (G16PC00016). Technical Liaisons from the USGS directed the survey planning and conducted reviews of the preliminary field data.



Figure 2. The Shawnee Mtns. and Granite Springs Valley, Pershing County.

Michael Hobbs, VP of Operations for EDCON-PRJ, managed the data acquisition, quality control, and communications with the USGS and DOE Technical Liaisons. Nick Anderson conducted the in-office quality control and processing of the data for EDCON-PRJ as well as the generation of the grids and databases used for daily reviews.

A large number of individuals contributed their time, talent, focus and effort to the field acquisition of the GeoDAWN data. Cloudstreet Flying Service, owned and operated by Joe Nance, conducted

the fixed-wing operations. Pilots and crew for Cloudstreet included Tracie Porter, Kyle Kendall, David Cyr, Bronson Macdonald, Nick Koukoutsakis, and Jim Witler.

Precision GeoSurveys, owned and operated by Harmen Keyser, conducted the helicopter acquisition over the Tonopah Block. Pilots and crew members for Precision included Ryan Snow, Bruce Larsen, Colin Pelton, Shawn Walker, Jenny Poon and Wendell Huttema.

Survey Planning

The first set of preliminary boundaries for the GeoDAWN Project was forwarded by the USGS staff to NV5 Geospatial in late September of 2020. The RFP and an initial Task Order was issued on November 10, 2020 and a Technical Exchange was conducted on November 20, 2020.

During the Technical Exchange with USGS and DOE scientists and staff, agreements were reached on the type of aircraft to be utilized, the use and characteristics of hybrid drape surfaces, and with excluding areas over Tribal Lands within the survey. The NV5 Team submitted its final Technical Proposal in January of 2021 and the Notice to Proceed was received in March of 2021.

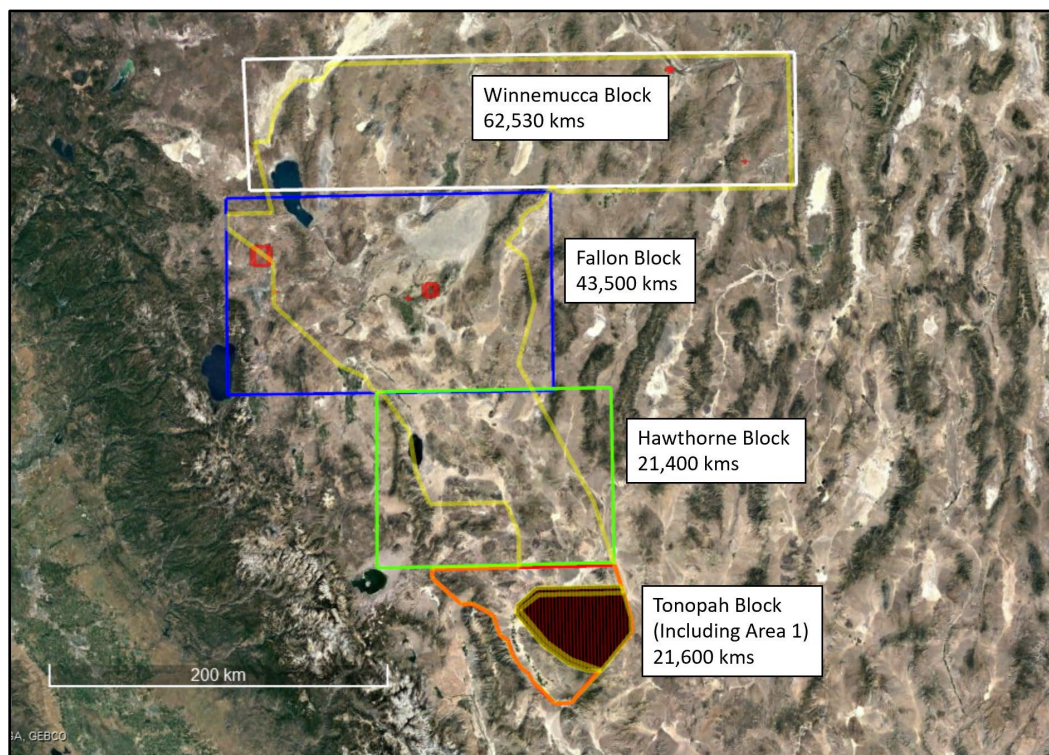


Figure 3. The Four GeoDAWN Acquisition Blocks.

Given the extremely large size of the project, it was decided to divide the survey into four separate acquisition blocks, each with an independent base of operations at the towns which became the name for each block. Acquisition was planned to begin at the northernmost block in the fall of

2021 and proceed south as weather permitted and then to return in 2022 to complete the unfinished blocks.

The USGS staff coordinated efforts to contact Tribes within the survey area for permission to overfly their lands during survey operations. Many tribes agreed to having the survey include their lands, but several did not. Blocks of excluded airspace were generated to honor this request and were incorporated into the flight navigation planning for the aircraft. The largest of these areas can be seen as red polygons in the image above (*Figure 3.*).

The GeoDAWN Project included two primary areas. Area 1 is located southwest of Tonopah and was incorporated into the Tonopah acquisition block. Area 1 (The Clayton Valley) was designated to be a high-density survey with primary flight traverse lines flown on a N 90° E heading at 200-meter spacings and tie lines to be flown on a N180°E heading at 2,000-meter spacings with a hybrid, pre-planned drape of 100 meters over basins and 150 meters over mountain ranges (*Figure 4.*).

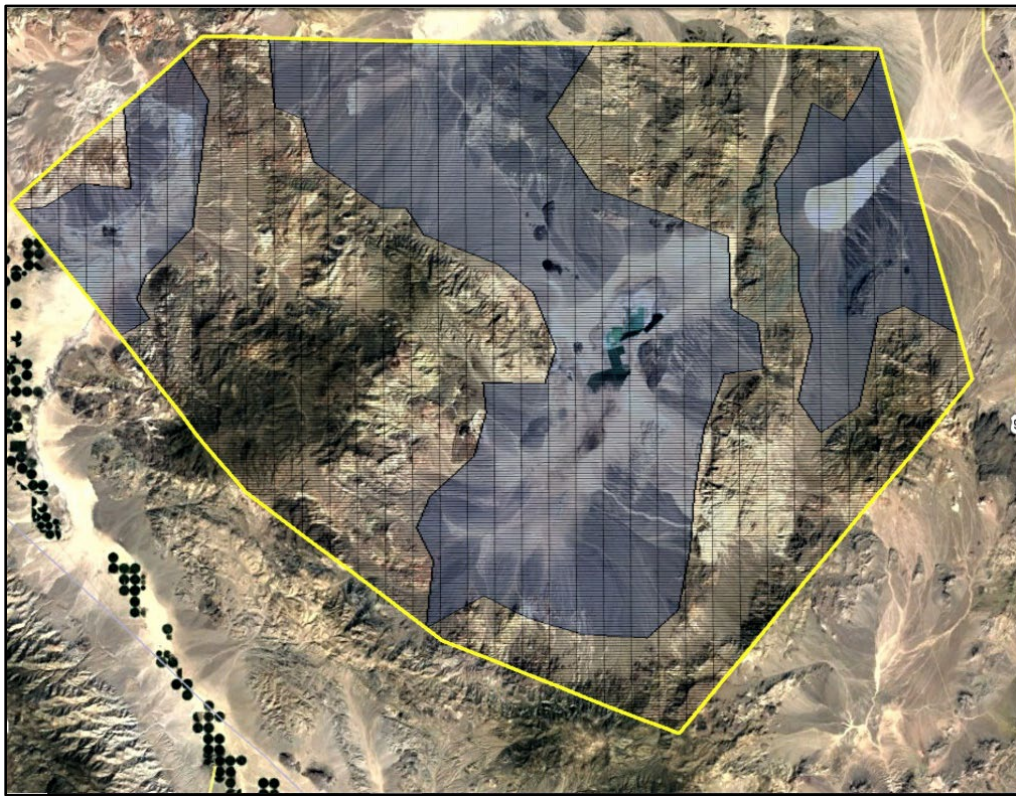


Figure 4. The Layout for Area 1 (Clayton Valley) with 100-meter drape areas shown in gray.

The rest of the GeoDAWN survey area (Area 2) was designated to have primary flight traverse lines flown on a N 90° E heading at 400-meter spacings and tie lines flown on a N180°E heading at 4,000-meter spacings, with a hybrid, pre-planned drape of 150 meters over the basins and 200 meters over the mountain ranges. Minor adjustments to the survey areas were made in consultation

with the USGS Technical Liaisons to square up boundary lines, improve the drape surface in areas of high relief, and eliminate short line segments without tie line intersections.

The pre-planned drape surfaces were generated using proprietary software that incorporates the digital terrain model, the required drape height, and the flight characteristics of the survey aircraft (survey speed, vertical acceleration, and rate of climb) to produce a hyperbolically-smoothed digital surface that allows close terrain following within a safe-flight envelope for the survey aircraft. A climb/dive angle of 22 degrees was used for both Area 1 and Area 2 flight planning.

Data Acquisition

Data acquisition for the GeoDAWN Project required 142 days of fixed-wing flights and 22 days of helicopter flights. A total of 149,030 kilometers of magnetic and radiometric data was acquired over 1,055 primary flight lines and 83 tie lines (**Figure 5**).

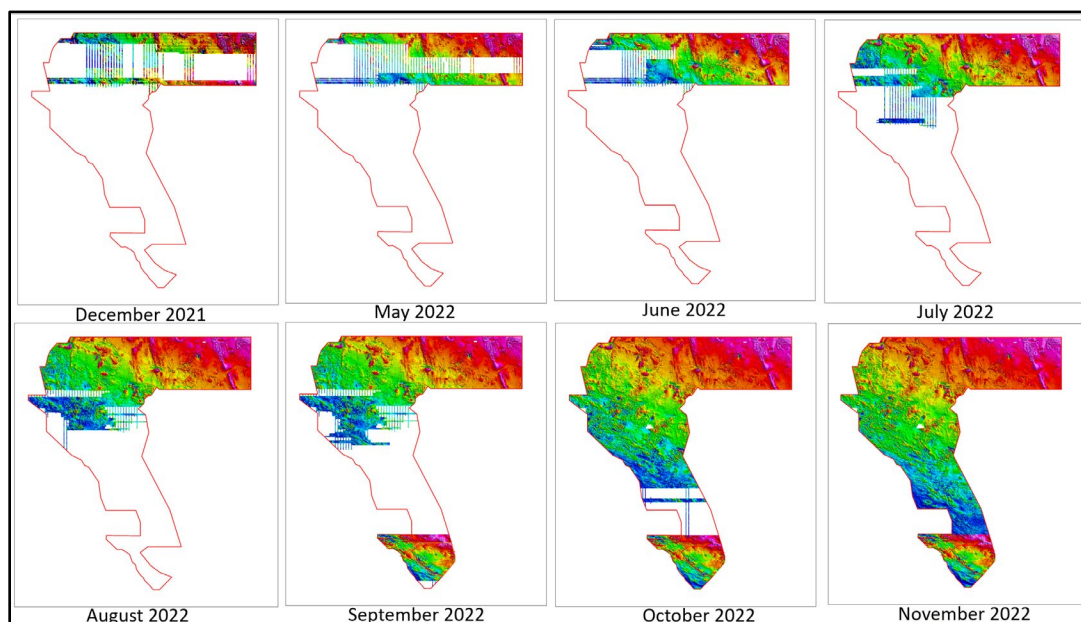


Figure 5. Magnetic Intensity Maps Showing Monthly Acquisition Progress.

Survey operations were originally planned to begin on August 1, 2021, but numerous large wildfires in the Sierra Nevada Mountains created unsafe haze conditions that persisted throughout the months of August, September, and much of October (**Figure 6**).

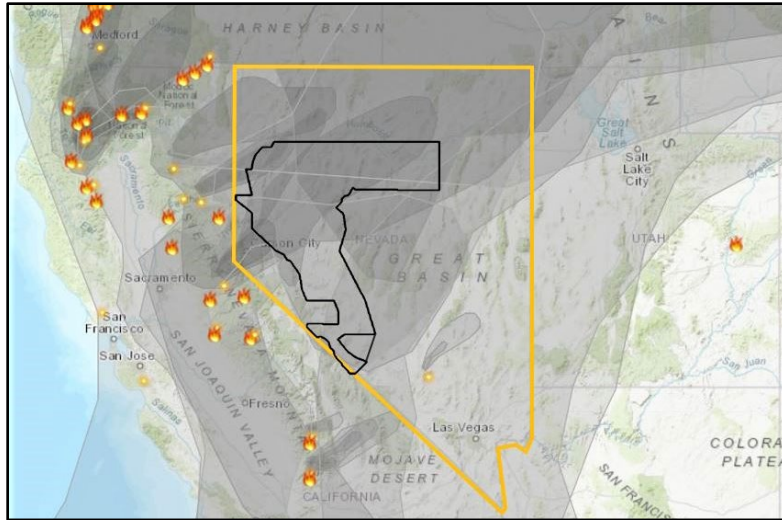


Figure 6. The NOAA Air Quality Map for August 26, 2021.

The wildfires finally abated mid-October, and the Cessna-180 fixed-wing crew mobilized to Winnemucca on October 30 and established the initial base of operations at the Winnemucca Municipal Airport. After conducting compensation flights, equipment calibrations and testing, and the establishment of a radiometric control line, data acquisition commenced on November 1 and continued until December 7, when an early, heavy snow blanketed the entire GeoDAWN survey area.

The crew was not able to reposition to the southern portion of the GeoDAWN project because heavy snow had collected on the higher ranges even 300 kilometers to the south. As a result, acquisition was halted for the balance of the year and the crew mobilized to the Salton Sea area and acquired data for the GeoFLIGHT project during the winter. A total of 21,546 line-kilometers of magnetic and radiometric data within the Winnemucca Block was acquired during the fall of 2021.

The Cloudstreet crew re-mobilized to Winnemucca on May 14, 2022 with a different aircraft, a Cessna Turbo 206, and re-commenced acquisition on May 15. They completed the 62,530 kilometers of the Winnemucca Block in July and relocated their base of operations to the Fallon Municipal Airport. At each of these operational bases, a new radiometric control line was established and was flown each day prior to data collection.



Figure 7. The Phoenix Mine, Lander County, Nevada.

On August 17, during a routine oil change and inspection of the Cessna 206, metallic fragments were discovered in the oil pan, necessitating a complete engine replacement. A new engine was located and shipped to Fallon in three weeks. The installation and testing of the new engine took another two weeks, and the Cessna 206 crew wasn't back on survey until September 26.

When the engine failure in the Cessna 206 occurred, it became clear that another aircraft would need to be brought in to assist with the remaining survey blocks in order to finish GeoDAWN in the fall of 2022. The decision was made to bring in Precision GeoSurvey's Bell Jet Ranger helicopter to acquire the Tonopah Block, the southernmost block and the one that included Area 1 (Clayton Valley). This approach was presented to the USGS and DOE Technical Liaisons on August 22 and met with their approval.



Figure 8. Installation of the Radar Altimeter at the Tonopah Municipal Airport.

The Precision Jet Ranger Crew arrived in Tonopah on September 12, installed surveying equipment the next day, and began data acquisition on September 14. They completed the 21,600 kilometers of the Tonopah Block, including Area 1, on October 6, having conducted 22 days of multiple sorties over 24 calendar days. The survey layout and pre-planned drape surface used for the Winnemucca, Fallon, and Hawthorne Blocks were used for the Precision Jet Ranger acquisition of the Tonopah Block. As with the other acquisition blocks, tie lines from one block extended a minimum of two kilometers into the adjacent block to ensure seamless merging of the datasets.



Figure 9. The Fallon Naval Air Station, Churchill County.

Once data acquisition resumed with the new engine, the Cessna 206 crew maintained excellent production, completing the remaining 47,219 kilometers of the Fallon and Hawthorne Blocks and completing the GeoDAWN Project on November 20. During this period they were able to complete the portions of the survey within the controlled airspace of the Fallon Naval Air Station (*Figure 9.*) and the restricted bombing ranges that could only be accessed on weekends.



Figure 10. Sand Mountain, Churchill County, Nevada.

Survey Methodology

Aircraft Compensations:

Prior to the commencement of survey operations, magnetic compensation and calibration routines were conducted to generate coefficients used during processing of the survey data. The pilots climbed to approx. 1,500 meters (4,900 ft.) above terrain and flew a square pattern in the four cardinal directions at a high altitude to suppress geologic effects on the readings. The pattern was flown twice, once “flat” as if conducting survey operations and once while performing moderate roll, pitch, and yaw maneuvers.

Several compensation routines were conducted during the GeoDAWN Project: One with the Cessna-180 at the beginning of survey operations, one with the Cessna-206 when it arrived at Winnemucca and another when it was fitted with a new engine, and one was conducted with the Jet Ranger helicopter when it began operations on the Tonopah Block. The data from the flux-gate magnetometers were processed to generate X, Y and Z-axis coefficients used to compensate for heading and maneuver error during survey operations.

Radar Altimeter Calibrations:

Radar altimeter calibrations were performed on each aircraft to calculate the coefficients to convert mVolts as supplied by the on-board altimeter to meters for the database. Identical FreeFlight Systems TRA3000 radar altimeters were mounted on all aircraft utilized on the GeoDAWN Project. Each radar altimeter calibration included coefficients generated at a range of altitudes from 60 meters to 450 meters.

Cosmic Calibration:

Radiometric data were acquired during the compensation routines for the purpose of determining the cosmic background and aircraft background counts to be used in the processing of spectrometer data. The cosmic background data were acquired for 20 minutes at an elevation of 1,500 meters (4,900 ft.) AGL for each aircraft.

Sensitivity and Height Attenuation Calibration:

The radiometric sensitivity and attenuation calibrations were conducted in conjunction with the radar altimeter calibrations. Data were collected during a series of passes over the same traverse at altitudes ranging from 60 meters to 450 meters to determine the height attenuation coefficients.

Daily Test Line Survey:

Radiometric control lines were established in proximity to each of the bases of operation. Data were collected at survey drape height each day prior to data acquisition. A minimum of 200 seconds of data were recorded each traverse. The control line data is used to correct for daily variations in temperature, humidity, and ground-level radon concentrations. A spreadsheet of the daily control line data was included for review in the materials uploaded daily to the USGS and is included in its entirety with the deliverables for this project (*Appendix B.*).

Daily Spectrometer Resolution Tests:

Thorium resolution tests were conducted each day prior to takeoff. First, the radiometric system was allowed to warm up for 10 minutes, then 5 minutes of data were recorded in a defined location, followed by 5 minutes of data collected with a Thorium source placed beneath the aircraft. A spreadsheet of these daily sensitivity tests was included in the materials uploaded daily to the USGS and is included with the deliverables for this project (*Appendix C.*).

Calculated Radar and GPS Position Corrections:

Instead of providing data from a barometric altimeter and in addition to radar altimeter coverage, we acquired and provide “calculated radar” elevations for the aircraft. The calculated radar values are generated by taking the NAD83/NAVD88 elevation values and subtracting the corresponding values from the USGS 3DEP Elevation.

We applied Precise Point Positioning corrections to the on-board GPS data, which results in GPS elevation accuracies at the sub-meter level. We utilize Inertial Explorer software, developed by Novatel, Inc., to accomplish the post-flight GPS corrections. Precise Point Positioning is similar to differential corrections made with a GPS base station and is much more accurate when flight line lengths are as long as were encountered in this survey.

Quality Control Measures:

The data from each flight were reviewed three times by separate individuals. The data were first reviewed by the pilot and crew immediately after each flight. Then the data were downloaded from the onboard computer and uploaded via ftp site to EDCON-PRJ’s offices in Lakewood, Colorado for formal quality review.

As a second check, Michael Hobbs, Operations Manager for EDCON-PRJ, would then download the data and review for quality and completeness. Finally, in Lakewood, each flight’s data were checked for quality by Nick Anderson, Airborne Data Processor for EDCON-PRJ, and then compensated and corrected for diurnal activity using data collected from the base station magnetometer. The data were reviewed for flight line accuracy, GPS resolution, radar altimeter positioning, magnetometer noise levels, and spectrometer count levels, tuning, and resolution.

Daily acquisition updates were forwarded via e-mail to the USGS and DOE staff. Two Geosoft databases, one for magnetic data and one for radiometric data, were updated daily and posted to the USGS folder on EDCON-PRJ's ftp site for review.

The data quality was very high during all phases of this survey. Diurnal activity during the survey varied, as would be expected given that acquisition took place over a 12-month period. All line segments re-flown for diurnal activity were acquired across at least two tie lines to ensure seamless line patches.



Figure 11. Star Peak, Humboldt Range, Pershing County, Nevada.

Safety Plan

All flight operations were conducted with safety and environmental awareness as the number one priority.



Figure 12. The EDCON-PRJ IAGSA Certificate.

EDCON-PRJ is a member of the International Airborne Geophysics Safety Association and subscribes to its safety procedures and policies. Our safety plan for each project includes:

1. A thorough review of the survey area by staff and pilots for risk assessment, drupe considerations, and public safety.
2. Daily inspection of equipment and aircraft for performance and operational safety.
3. Daily flight and safety planning were conducted with the crew prior to takeoff.
4. Real-time on-board weather radar and surface conditions reporting through satellite service.
5. An Emergency Response Plan (EMP) was completed prior to the commencement of survey operations and included contact information for pilot and crew, local response teams and authorities. Local authorities were notified of the location and extent of survey operations prior to their commencement.
6. Real time tracking of the survey aircraft and the chase vehicle through a satellite location service was conducted throughout survey operations.

Prior to each flight, the pilot performed a complete pre-flight inspection of the aircraft, including control surfaces, fuel contamination testing, communications equipment, geophysical equipment, and electrical systems. The pilot and crew then reviewed the flight plan for the day and discussed any potential safety issues due to terrain, obstacles, wind, and weather.

Equipment

Survey Aircraft:

Cloudstreet Flying Service employed both a Cessna C-180 (Reg. N-8814X) and C-206 Turbo (Reg. N-283TB) aircraft for the fixed-wing portions of the GeoDAWN Project (**Figure 13.**). EDCON-PRJ has a 17-year working history with Cloudstreet Flying Service, having jointly completed over 130 similar fixed-wing magnetometer projects throughout the United States, including numerous projects for the USGS.



Figure 13. The Cloudstreet Cessna 206 at the Fallon Airport.

The aircraft have STOL modification kits, which gives them better stability, climb and dive, and control characteristics while on survey operations. The fiberglass tail stinger, which measures 3.5 meters from aircraft tail to magnetometer sensor pod, isolates the sensor from vibration and electromagnetic noise sources.

The Bell Jet Ranger helicopter (Reg. C-FZHK) (**Figure 14.**) that was utilized for the Tonopah Block is owned and operated by Precision GeoSurveys, Inc. of Langley, British Columbia. Precision and EDCON-PRJ have been partnering for the past year to increase production on large EarthMRI surveys. Precision also provided fixed-wing assistance early in 2022 on the EarthMRI GeoFLIGHT project in southern California and is currently conducting the acquisition for the EarthMRI Republic Survey in northeast Washington State.



Figure 14. The Bell Jet Ranger at a refueling stop on the Tonopah Block.

Airborne Magnetometers:

A Geometrics G823A Magnetometer was installed in the aft section of the tail stinger of the Fixed-wing aircraft. This Cesium-vapor magnetometer has a sensitivity of 0.015 nT over a range of 20,000 – 100,000 nT, with a sensor noise level of less than 0.002 nT. The airborne magnetic field data were recorded at 10 Hz .



Figure 15. Pilot's view of the Gold Zone Mine, Nye County, Nevada.

A Scintrex CS-3 split-beam cesium vapor magnetometer was installed in the sensor pod of the forward stinger on the Bell Jet Ranger (**Figure 15.**). The Scintrex CS-3 has a sensitivity of 0.0006 nT over a range of 15,000 – 105,000 nT, with a noise level of 0.002 nT. The airborne magnetic field data were recorded at 10 Hz . The sensor was oriented vertically during survey operations.

Gamma Ray Spectrometers:

Nuvia Dynamics Advanced Gamma Ray Spectrometers (AGRSs) were utilized by both the fixed-wing and helicopter platforms for the GeoDAWN Project (**Figure 16.**). The fixed-wing system consisted of 16.8 liters of downward-looking NaI detectors and a 2.1 liter upward-looking detector. The helicopter system consisted of 16.8 liters of downward-looking NaI detectors and a 4.2 liter upward-looking detector. The AGRS systems feature a 512-channel microprocessor, automatic calibration, and tuning using natural background elements.

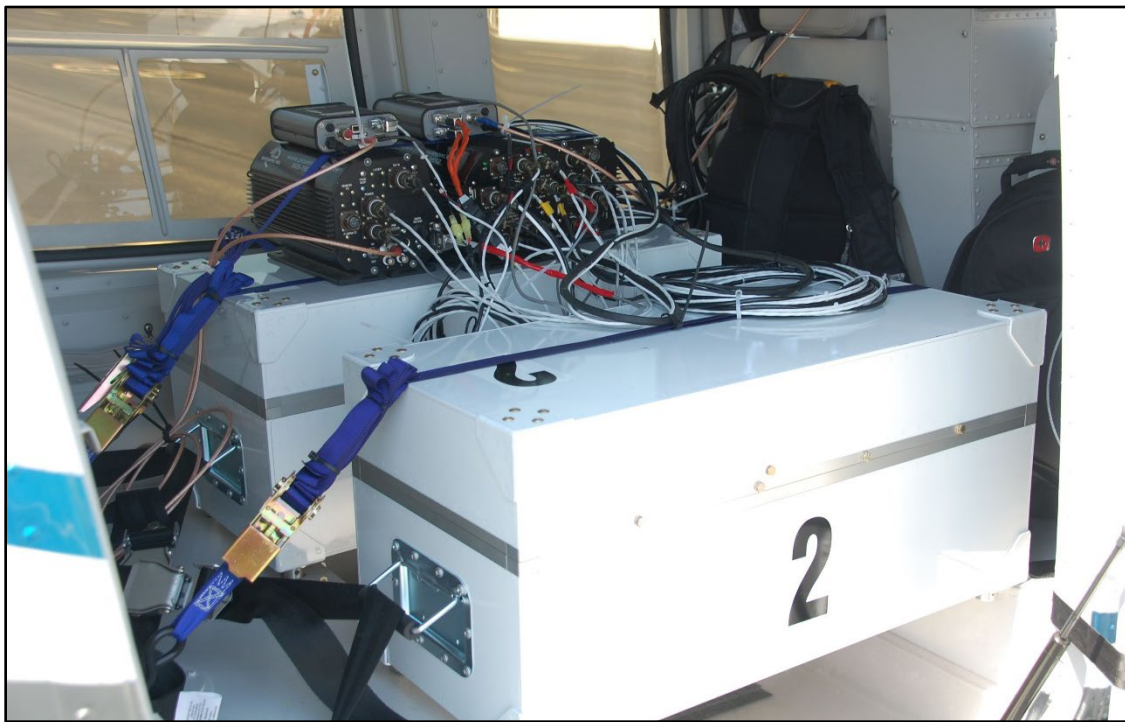


Figure 16. The Nuvia Dynamics Advanced Gamma Ray Spectrometer System.

The spectrometers were calibrated prior to the start of survey operations and were tested for sensitivity and resolution prior to each day's flights. Data was logged over a control line each morning enroute to the survey area. The daily and control line data and resolution tests are attached to this report as (**Appendix B.**) and (**Appendix C.**).

Global Positioning Systems:

Novatel PowerPak7 dual-band, geodetic-quality GPS receivers were installed in all aircraft and supplied navigation positioning as well as geophysical data points. WAAS DGPS corrections were utilized, providing sub-meter horizontal and vertical positioning (2D-RMS 95% probability). The Novatel Receiver has a proven accuracy using WAAS of +/- 0.6 m. GPS data were recorded at 5 Hz during all survey operations. Precise Point Positioning (PPP) corrections were employed post-flight to increase positional accuracy even further.

Fluxgate Compensation Magnetometers:

An Applied Physics Systems Model 539 High Speed Digital 3-axis Fluxgate Magnetometer was used to provide compensation data to the fixed-wing aircraft during survey operations. The Model 539 records X, Y, and Z- axis data at 30 Hz.

A Billingsley Model TFM100G2 Triaxial Fluxgate Magnetometer provided compensation data to the Bell Jet Ranger during survey operations. The Model TFM 100G2 was recording data at 50 Hz.

Radar Altimeters:

FreeFlight Systems TRA-3000 radar altimeters were mounted on the bottom of each of the aircraft and provided ground clearance information to the pilot as well as aircraft elevation positions to the geophysical data stream. The TRA-3000 Radar Altimeter provides precise altitude information at 10 Hz, with a digital recording resolution of 0.25 ft.

Survey Navigation Systems:

An Ag-Nav Platinum three-dimensional survey and aircraft guidance system was utilized to provide pilot guidance for the fixed-wing aircraft. The system provides the pilot with cross-track X-Y guidance and a vertical lightbar supplies Z-axis positions relative to the pre-planned drape surface.

The Bell Jet Ranger utilized a Nuvia Dynamics Airborne Geophysical Information System (AGIS) for survey navigation. Three-axis guidance is supplied by a flat screen display mounted on the instrument dash of the helicopter. The display also allows the pilot to monitor the status of all on-board geophysical systems, so that a geophysical operator is not necessary.

Ground Based Instrumentation:

A Geometrics G-858 Cesium magnetometer was used as the primary base station during fixed-wing survey operations along with a Geometrics 856 magnetometer deployed as a backup system. Two GEM GSM-19T base station magnetometers were used during helicopter operations. The time-synchronized ground magnetic field data were digitally recorded at a 1.0 Hz and to a resolution of 0.1 nT. The base station locations were located in quiet areas distanced from electromagnetic noise sources and vehicle traffic at the four bases of operation (**Figure 17.**).

Best efforts were made to position the bases of operations to minimize areas more than the requested 80 kilometers from base station magnetometers. Figure 17 shows each base station location with an 80 kilometer range ring.

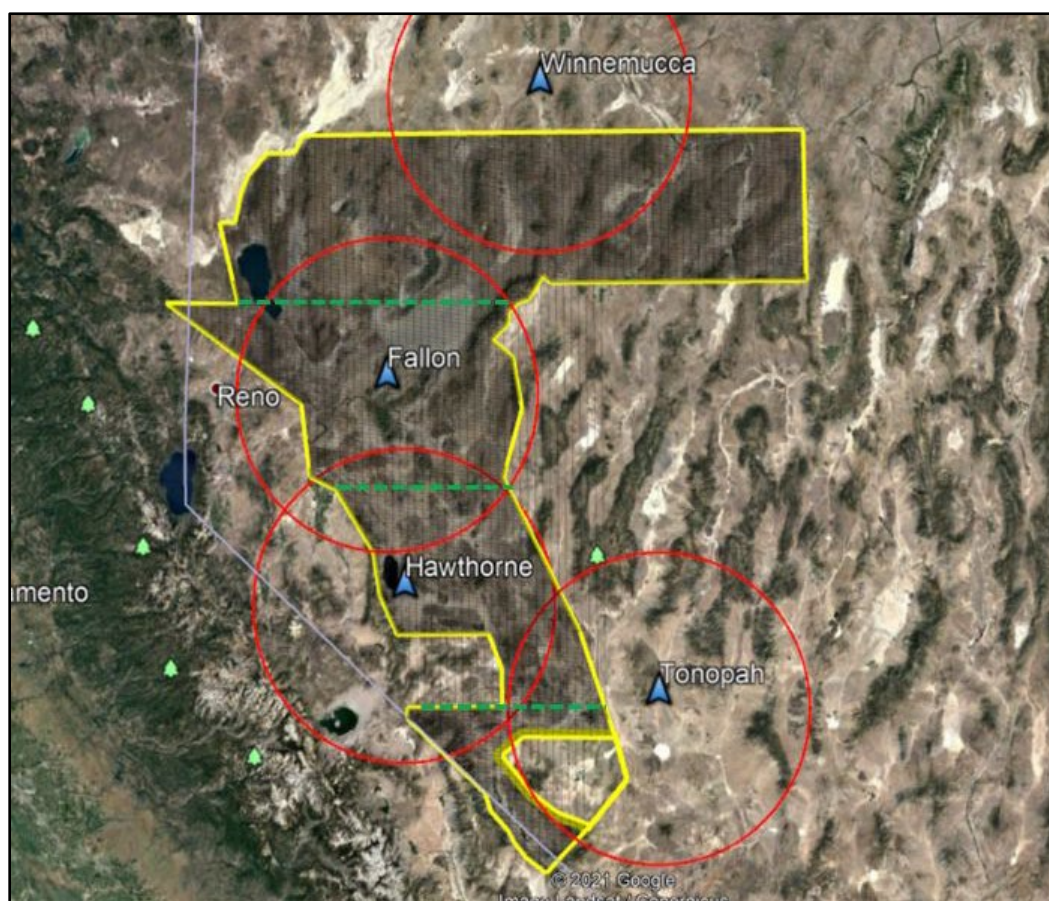


Figure 17. The Base Station Locations for the GeoDAWN Project.

Base Station Locations (UTM Zone 11 WGS84):

Winnemucca:	432658 mE	4528343 mN
Fallon:	349803 mE	4373452 mN
Hawthorne:	357908 mE	4266948 mN
Tonopah:	491299 mE	4213070 mN

Data Processing

A. Flight Path Recovery

The DGPS navigation vertical and horizontal coordinate outputs were recorded as latitude, longitude, and elevation using the WGS84 geographic coordinate system and vertical datum. WGS84 Vertical Datum Mapping parameters for processed digital and mapped data are shown below.

Projection Information:

Projection :	Geographic (Lat/Lon)
Ellipsoid:	WGS84
Datum :	WGS84

Projection:	UTM, Zone 11
Ellipsoid:	WGS84
Datum:	WGS84

A speed check on the location data was completed, and the line location with the derived aircraft speed information was mapped for editing. After editing, the GPS data were accepted for the final flight-path map production.

B. Magnetic Data

1. Preliminary Processing and Data Editing

- a. Digital magnetic and navigation data were transmitted from the field to EDCON-PRJ's data site for daily evaluation and verification.
- b. Profile plots of the magnetic data for each line were inspected for noisy or missing data.
- c. The data quality was good, and no filters were applied.
- d. De-culturing of the data was not conducted.

2. I.G.R.F.

The International Geomagnetic Reference Field (IGRF 2020), updated to the dates of the survey, was calculated for the survey then removed for the diurnal correction and added back into diurnal corrected magnetics before leveling. It was removed permanently after leveling per the USGS request.

3. *Diurnal Correction*

The base magnetometer data were inspected and compared with the observed magnetic data trace. The following diurnal correction, in addition to the removal of diurnal by the line adjustment procedures, was applied to the dataset:

The observed diurnal data were first low-pass filtered to remove noise and then the IGRF was removed from each base location and subtracted from the IGRF corrected observed magnetic data.

Winnemucca IGRF value:	50195 – Flight1 to 87(fixed wing)
Fallon IGRF Value:	49107 – Flight 88 to 148(fixed wing)
Hawthorne IGRF Value:	48566 – Flight 149 to 161(fixed wing)
Tonopah IGRF Value:	48566 – all flights (helicopter)

4. *Leveling and Micro-leveling*

Mis-ties at line intersections were calculated and adjusted to minimize mis-tie errors. Initial leveling adjustments were completed using a DC level adjustment to compensate for long wavelength diurnal effects. The average intersection mis-tie before DC adjustment was 13.82 nT; after DC adjustment, the average mis-tie was 3.12 nT. After micro-leveling the average mis-tie was 0.57 nT.

After DC-leveling, a slight strike filter was applied to the magnetic grid to remove any line-pulls, which was then used as a leveling surface in micro-leveling.

The DC-leveled data showed line some line-oriented anomalies. A strike filter removed these minor line pulls. Then, the DC-leveled data were subtracted from the strike filtered data to make a difference channel. A new leveled magnetics channel was created by subtracting the filtered difference from the DC-leveled mag. This non-linear adjustment helped to maintain valid detail with minimal filtering.

After micro-leveling a small number of tie-lines still were not at the correct level due to lower elevations than the corresponding flight lines. As a result, the tie-lines in these locations showed anomalies that were not present on the flight lines. To get these intersections leveled, the tie-lines were forced to the flight-line levels by linear interpolation, then micro-leveled again, which removed anomalies caused by the linear interpolation.

After gridding the leveled data, a 150-300 bandpass filter was applied to the tmi to reduce the sharpness of the peaks and to improve data quality to any operator/filters applied to the grids.

5. Reduction to the Pole

Reduction-to-the pole calculates the field that would be observed if the survey area were located at the North magnetic pole. This transformation shifts the magnetic anomalies more nearly over the causative bodies.

The reduced-to-pole grid for Area 1 was calculated using the following parameters:

Parameter	Value
* IGRF/DGRF:	IGRF
* Date required (yyyy/mm/dd):	2022/12/29
* IGRF Model Year (ignored for DGRF):	auto
* Longitude (-180 to 180):	-117.634238
* Latitude (-90 to 90):	37.778836
* Elevation (metres):	1531
Field Strength (nT):	48558.4657
Inclination (degrees):	62.16692855
Declination (degrees):	12.19259806

Figure 18. IGRF Values for Area 1.

The reduced-to-pole grid for Area 2 was calculated using the following parameters:

Parameter	Value
* IGRF/DGRF:	IGRF
* Date required (yyyy/mm/dd):	2022/12/09
* IGRF Model Year (ignored for DGRF):	auto
* Longitude (-180 to 180):	-118.15931
* Latitude (-90 to 90):	39.25151
* Elevation (metres):	2032
Field Strength (nT):	49292.1494
Inclination (degrees):	63.27245994
Declination (degrees):	12.61956080

Figure 19. IGRF Values for Area 2.

C. Spectrometer Data

Spectrometer data was processed using Geosoft Praga 4 software. The processing workflow followed Sections 1-6 below:

1. *Filtering*

All spectrometer data had NASVD spectra smoothing applied to it.

2. *Reduction to Standard Temperature and Pressure*

A temperature and relative humidity sensor was mounted in each aircraft and recorded data during all survey traverses. The radar altimeter data were reduced to effective height in meters using the temperature and pressure data millibars.

3. *Dead Time Correction*

The data were corrected for dead-time at 10 milliseconds.

4. *Aircraft, Cosmic, and Radon Background*

Test flights flown at 1,500 meters (4,900 feet) AGL were performed to calculate the background values per I.A.E.A. standard procedures. Radon was not constant during the period of data acquisition and was removed using the data collected by the upward crystal and comparing the downward uranium and thorium data to upward uranium and thorium (**Table 1**).

Table 1. Aircraft and Cosmic Background Values.

	Potassium	Thorium	Uranium	Total Count
Aircraft Background	4	1	0.5	25
Cosmic Background	0.1007	0.087	0.1074	1.82

Radon removal is one of the most difficult tasks to be done when processing radiometric data. Radon concentrations vary widely from day-to-day due to fluctuations in temperature, pressure, humidity and atmospheric inversions. Although utilizing counts from the upward-looking crystal are the preferred method of radon removal, the process is not perfect.

In several places within the northern portion of the GeoDAWN Survey, slight banding can be seen in the Uranium Concentration Grid (22103_u.grd). Data were collected over the course of a full year in the northern portion of the survey, making radon concentrations difficult to normalize. The resultant banding in the Uranium Concentration Grid could be removed by filtering, but the decision was made to not reduce the overall resolution of the radiometric data by eliminating this cosmetic effect.

5. Compton Stripping & Window Sensitivities

A series of pad calibrations were performed on April 7, 2022, at the Nuvia Dynamics Facilities in Mississauga, Ontario. The values obtained from the pad calibration were used to determine Stripping Coefficients and Window Sensitivities (**Tables 2,3.**).

Table 2. Stripping Coefficients

Alpha	0.3322
Beta	0.4552
Gamma	0.8937
A	0.05
B	0
G	0

Table 3. Window Sensitivities

Potassium (%K)	212
Uranium (ppm)	50
Thorium (ppm)	55
Total Count (nGy/h)	125

6. Attenuation Correction

The data were corrected for attenuation due to altitude. Calibration flights were performed with each aircraft along control lines at elevations from 30 meters – 455 meters (100 feet – 1500 feet) (**Table 4.**).

Table 4. Attenuation Corrections

Potassium/m@STP	Thorium/m@STP	Uranium/m@STP	Total Count/m@STP
-0.0025	-0.0023	-0.0024	-0.0025

7. Daily Resolution Test

The AGRS spectrometer is a self-calibrating system that calibrates itself to Thorium peaks as it logs data. A Thorium sensitivity test was conducted each day prior to takeoff. First, the radiometric system was allowed to warm up for 10 minutes, then five minutes of data were recorded in a defined location, followed by five minutes of data collected with a Thorium source placed beneath the aircraft. The data were checked each day to make sure the resolution of the spectra remained well within specification (*Appendix C*).

8. Daily Control Line

Radiometric control lines were established near each base of operations. Data were collected at survey drape height each day prior to data acquisition. A minimum of 200 seconds of data were recorded each traverse. Data were collected along this same traverse daily and checked for consistency (*Appendix B*).

Deliverables

A. Digital Data (each file has a suffix of _a1 for area1 data and _a2, for area2 including area1.)

- 22103_tmi.grd: Total Magnetic Intensity Grid in Geosoft Grid Format
- 22103_tmi_upcont_tmi150.grd: TMI Grid of Area 2 with Area 1 Upward Continued
- 22103_rtp.grd: Reduced To Pole TMI Grid in Geosoft Grid Format
- 22103_vg.grd: Vertical Gradient of the TMI Grid in Geosoft Grid Format
- 22103_hg.grd: Horizontal Gradient of the TMI Grid in Geosoft Grid Format
- 22103_dem.grd: Topography Grid in Geosoft Grid Format
- 22103_k.grd: Potassium Grid in Geosoft Grid Format
- 22103_tc.grd: Total Count Grid in Geosoft Grid Format
- 22103_th.grd: Thorium Grid in Geosoft Grid Format
- 22103_thk.grd: Thorium/Potassium ratio Grid in Geosoft Grid Format
- 22103_u.grd: Uranium Grid in Geosoft Grid Format
- 22103_uk.grd: Uranium/Potassium ratio Grid in Geosoft Grid Format
- 22103_uth.grd: Uranium/Thorium ratio Grid in Geosoft Grid Format
- 22103_ternary.map: Ternary image in Geotiff Map Format
- 22103_mag.gdb: Survey magnetic line data in Geosoft Database Format
- 22103_spec.gdb: Survey radiometric line data in Geosoft Database Format

Map Projection Information for contract Task Order 140G0221F0024:

Projection:	UTM
Datum:	WGS84
Base Latitude:	0
Central Meridian:	-117
False Easting:	500,000 meters
False Northing:	0 meters

Grid Information for Task Order 140G0221F0024:

Area 1:

Origin of Grid : 406600.0, 4153800.0
Grid Interval: 50 meters

Area 2:

Origin of Grid : 243200.0, 4135400.0
Grid Interval: 100 meters

Profile Data Channel Descriptions (Magnetics):

Channel Name	Description	Units
Line	Line Number	
flight	Flight Number	
heading	Heading	Degrees
fid	Fiducial	
time	Time	HH:MM:SS.SSS
date	Date	YYYY/MM/DD
lat	Latitude	Decimal Degrees, WGS84
lon	Longitude	Decimal Degrees, WGS84
x	UTM X Coordinate, Zone 11	Meters, WGS84
y	UTM Y Coordinate, Zone 11	Meters, WGS84
dem	Digital Elevation Model	Meters
drape	Drape Surface	Meters
gps_elev	Gps elevation	Meters (PPP corrected)
radar	Radar Altimeter	Meters
calc_radar	Calculated Radar (gps_elev – DEM)	Meters
base	Base Station Magnetics	nT
raw_mag	Raw magnetics	nT
comp_mag	Compensated Magnetics	nT
igrf_cor_base	IGRF correction for the Base	nT
filtered_base	Filtered Base Magnetics	nT
diurnally_cor_mag	(IGRF Corrected Comp_mag – IGRF corrected filt base)	nT
dc_level_mag	Diurnally corrected DC leveled Magnetics.	nT
micro_leveled_surface	Magnetics to Micro-level onto	nT
micro_leveled_correction	Micro-leveling Correction	nT
micro_Leveled_mag	Micro-leveled diurnally corrected magnetics	nT
igrf_correction	IGRF Correction	nT
final_Mag	Diurnally and IGRF Corrected and leveled Magnetics	nT

Profile Data Channel Descriptions (Radiometrics):

Channel Name	Description	Units
Line	Line Number	
flight	Flight Number	
heading	Heading	Degrees
fid	Fiducial	
date	Date	YYYY/MM/DD
time	Time	HH:MM:SS.SS
Lat	Latitude	Decimal Degrees, WGS84
Lon	Longitude	Decimal Degrees, WGS84
x	UTM X Coordinate, Zone 11	Meters, WGS84
y	UTM Y Coordinate, Zone 11	Meters, WGS84
dem	Digital Elevation Model	Meters
drape	Drape Surface	Meters
gps_elev	Gps elevation	Meters (PPP corrected)
radar	Radar Altimeter (Area 1 only)	Meters
calc_radar	Calculated Radar (gps_elev – DEM)	Meters
radar_stp	Radar corrected to STP	Meters
Dose	Dose	nGy/hour
cosmic_raw	Raw Cosmic Count	Cps
k_raw	Potassium Raw	Cps
tc_raw	Total Count Raw	Cps
Up_tc_raw	Upward Count Raw	Cps
th_raw	Thorium Raw	Cps
u_raw	Uranium Raw	Cps
kcorr	Corrected Potassium	%K
tccorr	Corrected Total Count	nGy/hour
thcorr	Corrected Thorium	Ppm
ucorr	Corrected Uranium	Ppm
thk_ratio	Thorium/Potassium Ratio	
uk_ratio	Uranium/Potassium Ratio	
uth_ratio	Uranium/Thorium Ratio	
ISPD	Downward Spectrum	
ISPU	Upward Spectrum	
NASVD_spectrum	Smoothed Downward Spectrum	
radon	Radon Correction	Cps
pressure	Air Pressure	Mbar
temp	Temperature	Degrees Celsius
humidity	Humidity	%

Appendix A: Daily Survey Log

<u>Date</u>	<u>Aircraft</u>	<u>Base</u>	<u>Comments</u>	<u>Kms</u>
10.30.21	Cessna-180	Winnemucca	Radar Calibration, Attenuation, Set Control Line	
10.31.21	Cessna-180	Winnemucca	Compensation, Cosmic Background	
11.01.21	Cessna-180	Winnemucca	7 Partial Flight Lines	720
11.02.21	Cessna-180	Winnemucca	1 Flight Line, 5 Partial Flight Lines	1035
11.03.21	Cessna-180	Winnemucca	1 Flight Line, 5 Partial Flight Lines	738
11.05.21	Cessna-180	Winnemucca	9 Partial Flight Lines	575
11.07.21	Cessna-180	Winnemucca	3 Partial Flight Lines, 1 Reflight Line	365
11.08.21	Cessna-180	Winnemucca	6 Partial Flight Lines, 2 Partial Tie Lines	585
11.11.21	Cessna-180	Winnemucca	2 Partial Flight Lines, 7 Partial Tie Lines	633
11.12.21	Cessna-180	Winnemucca	1 Flight Line, 6 Partial Flight Lines, 5 Partial Tie Lines	1309
11.13.21	Cessna-180	Winnemucca	9 Partial Flight Lines, 3 Partial Tie Lines	1081
11.14.21	Cessna-180	Winnemucca	11 Partial Flight Lines, 4 Tie Lines	1355
11.17.21	Cessna-180	Winnemucca	1 Partial Flight Line	145
11.20.21	Cessna-180	Winnemucca	10 Partial Flight Lines	1225
11.21.21	Cessna-180	Winnemucca	6 Partial Flight Lines, 4 Partial Tie Lines	1030
11.22.21	Cessna-180	Winnemucca	6 Partial Flight Lines	595
11.25.21	Cessna-180	Winnemucca	6 Partial Flight Lines, 2 Partial Tie Lines	895
11.26.21	Cessna-180	Winnemucca	9 Partial Flight Lines	1010
11.27.21	Cessna-180	Winnemucca	9 Partial Flight Lines, 3 Tie Lines	1190
11.28.21	Cessna-180	Winnemucca	1 Flight Line, 10 Partial Flight Lines, 1 Reflight Line, 2 Tie Lines	1185
11.30.21	Cessna-180	Winnemucca	11 Partial Flight Lines, 2 Tie Lines, 1 Partial Tie Line	1093
12.01.21	Cessna-180	Winnemucca	6 Partial Flight Lines	634
12.02.21	Cessna-180	Winnemucca	9 Partial Flight Lines	1152
12.03.21	Cessna-180	Winnemucca	10 Partial Flight Lines	1384
12.04.21	Cessna-180	Winnemucca	6 Partial Flight Lines	632
12.07.21	Cessna-180	Winnemucca	9 Partial Flight Lines	980
12.14.21	Cessna-180	Winnemucca	Survey operations ended for year due to snowpack.	
5.14.22	Cessna-206	Winnemucca	Compensation, Cosmic Background, Radar Calibration, Attenuation	
5.15.22	Cessna-206	Winnemucca	2 Partial Flight Lines, 2 Tie Lines	437
5.16.22	Cessna-206	Winnemucca	4 Partial Flight Lines, 2 Tie Lines	697
5.17.22	Cessna-206	Winnemucca	7 Partial Flight Lines, 2 Tie Lines	1430
5.18.22	Cessna-206	Winnemucca	5 Partial Flight Lines, 2 Partial Tie Lines	884
5.21.22	Cessna-206	Winnemucca	8 Partial Flight Lines, 3 Partial Tie Lines	1401
5.22.22	Cessna-206	Winnemucca	4 Partial Flight Lines, 3 Tie Lines	880
5.23.22	Cessna-206	Winnemucca	4 Flight Lines, 1 Tie Line	925
5.24.22	Cessna-206	Winnemucca	4 Partial Flight Lines	412
5.25.22	Cessna-206	Winnemucca	11 Partial Flight Lines, 3 Tie Lines	1479
5.26.22	Cessna-206	Winnemucca	6 Partial Flight Lines	442
5.27.22	Cessna-206	Winnemucca	4 Partial Flight Lines	352
5.31.22	Cessna-206	Winnemucca	13 Partial Flight Lines	1250
6.01.22	Cessna-206	Winnemucca	9 Partial Flight Lines	1319

<u>Date</u>	<u>Aircraft</u>	<u>Base</u>	<u>Comments</u>	<u>Kms</u>
6.02.22	Cessna-206	Winnemucca	7 Partial Flight Lines	1026
6.03.22	Cessna-206	Winnemucca	4 Partial Flight Lines	670
6.06.22	Cessna-206	Winnemucca	6 Partial Flight Lines	912
6.07.22	Cessna-206	Winnemucca	6 Partial Flight Lines	1024
6.08.22	Cessna-206	Winnemucca	8 Partial Flight Lines	1536
6.09.22	Cessna-206	Winnemucca	5 Partial Flight Lines	637
6.10.22	Cessna-206	Winnemucca	5 Partial Flight Lines	699
6.14.22	Cessna-206	Winnemucca	4 Partial Flight Lines	928
6.15.22	Cessna-206	Winnemucca	4 Partial Flight Lines, 10 Tie Lines	1175
6.16.22	Cessna-206	Winnemucca	4 Partial Flight Lines	592
6.20.22	Cessna-206	Winnemucca	5 Partial Flight Lines, 7 Tie Lines	1307
6.22.22	Cessna-206	Winnemucca	6 Partial Flight Lines	1106
6.24.22	Cessna-206	Winnemucca	11 Partial Flight Lines	690
6.26.22	Cessna-206	Winnemucca	4 Partial Flight Lines	462
6.27.22	Cessna-206	Winnemucca	7 Partial Flight Lines	850
6.29.22	Cessna-206	Winnemucca	9 Partial Flight Lines	711
6.30.22	Cessna-206	Winnemucca	6 Partial Flight Lines	714
7.01.22	Cessna-206	Winnemucca	3 Partial Flight Lines	223
7.02.22	Cessna-206	Winnemucca	16 Partial Flight Lines	120
7.03.22	Cessna-206	Winnemucca	10 Partial Flight Lines, 1 Partial Tie Line	446
7.04.22	Cessna-206	Winnemucca	7 Partial Flight Lines, 2 Partial Tie Lines	862
7.05.22	Cessna-206	Winnemucca	9 Partial Flight Lines	882
7.06.22	Cessna-206	Winnemucca	10 Partial Flight Lines	844
7.07.22	Cessna-206	Winnemucca	10 Partial Flight Lines	611
7.08.22	Cessna-206	Winnemucca	9 Partial Flight Lines	700
7.09.22	Cessna-206	Winnemucca	3 Partial Flight Lines, 2 Partial Tie Lines	486
7.10.22	Cessna-206	Winnemucca	8 Partial Tie Lines	720
	C-206 Crew Repositioned to Fallon			
7.11.22	Cessna-206	Fallon	4 Partial Flight Lines, Set New Control Line	360
7.14.22	Cessna-206	Fallon	13 Partial Flight Lines, 2 Partial Tie Lines	956
7.15.22	Cessna-206	Fallon	16 Partial Flight Lines	1291
7.16.22	Cessna-206	Fallon	10 Partial Flight Lines, 2 Partial Tie Lines	1012
7.17.22	Cessna-206	Fallon	8 Partial Flight Lines, 2 Partial Tie Lines	896
7.20.22	Cessna-206	Fallon	14 Partial Flight Lines, 2 Partial Tie Lines	1108
7.21.22	Cessna-206	Fallon	13 Partial Flight Lines, 2 Partial Tie Lines	1269
7.23.22	Cessna-206	Fallon	18 Partial Flight Lines, 2 Partial Tie Lines	1389
7.24.22	Cessna-206	Fallon	14 Partial Flight Lines, 4 Partial Tie Lines	1236
7.31.22	Cessna-206	Fallon	15 Partial Flight Lines	973
8.01.22	Cessna-206	Fallon	20 Partial Flight Lines, 1 Partial Tie Line	1233
8.02.22	Cessna-206	Fallon	17 Partial Flight Lines, 2 Partial Flight Lines	1241
8.03.22	Cessna-206	Fallon	22 Partial Flight Lines	1344
8.04.22	Cessna-206	Fallon	17 Partial Flight Lines, 6 Partial Tie Lines	967
8.05.22	Cessna-206	Fallon	6 Partial Flight Lines	446
8.06.22	Cessna-206	Fallon	9 Partial Flight Lines	328
8.07.22	Cessna-206	Fallon	14 Partial Flight Lines, 1 Tie Line	1524

<u>Date</u>	<u>Aircraft</u>	<u>Base</u>	<u>Comments</u>	<u>Kms</u>
8.08.22	Cessna-206	Fallon	14 Partial Flight Lines, 3 Partial Tie Lines	653
8.09.22	Cessna-206	Fallon	15 Partial Flight Lines	884
8.10.22	Cessna-206	Fallon	7 Partial Flight Lines, 8 Tie Lines	1012
8.11.22	Cessna-206	Fallon	16 Partial Flight Lines	1441
8.12.22	Cessna-206	Fallon	10 Partial Flight Lines	972
8.13.22	Cessna-206	Fallon	12 Partial Flight Lines	1234
8.14.22	Cessna-206	Fallon	11 Partial Flight Lines	1181
8.17.22	Cessna-206	Fallon	8 Partial Flight Lines	872
	Cessna Flights Suspended for Replacement of Aircraft Engine			
9.12.22	Bell Jet Ranger	Tonopah	Compensation, Cosmic Background	
9.13.22	Bell Jet Ranger	Tonopah	2 Tie Lines as Test Data	
9.14.22	Bell Jet Ranger	Tonopah	6 Flight Lines	877
9.15.22	Bell Jet Ranger	Tonopah	8 Flight Lines, 4 Tie Lines	1100
9.16.22	Bell Jet Ranger	Tonopah	12 Flight Lines	1265
9.17.22	Bell Jet Ranger	Tonopah	6 Flight Lines	615
9.19.22	Bell Jet Ranger	Tonopah	8 Flight Lines, 2 Partial Flight Lines	824
9.20.22	Bell Jet Ranger	Tonopah	6 Flight Lines, 2 Partial Flight Lines	691
9.21.22	Bell Jet Ranger	Tonopah	16 Flight Lines, Radar Calibration, Attenuation	1406
9.22.22	Bell Jet Ranger	Tonopah	20 Flight Lines	1590
9.23.22	Bell Jet Ranger	Tonopah	8 Flight Lines	622
9.26.22	Bell Jet Ranger	Tonopah	18 Flight Lines	1333
	Cessna Flights Recommence with New Engine			
	Cessna-206	Fallon	17 Partial Flight Lines, Compensation	324
9.27.22	Bell Jet Ranger	Tonopah	14 Flight Lines	912
	Cessna-206	Fallon	24 Partial Flight Lines	1088
9.28.22	Bell Jet Ranger	Tonopah	18 Flight Lines	1020
	Cessna-206	Fallon	5 Partial Flight Lines, 2 Partial Tie Lines	563
9.29.22	Bell Jet Ranger	Tonopah	26 Flight Lines	1160
	Cessna-206	Fallon	17 Partial Flight Lines, 6 Partial Tie Lines	1010
9.30.22	Bell Jet Ranger	Tonopah	30 Flight Lines, 9 Tie Lines	1210
	Cessna-206	Fallon	34 Partial Flight Lines, 2 Partial Tie Lines	1086
10.01.22	Bell Jet Ranger	Tonopah	6 Flight Lines, 16 Tie Lines (Begins Area 1 In-Fill)	1145
	Cessna-206	Fallon	13 Partial Flight Lines, 8 Tie Lines	971
10.02.22	Bell Jet Ranger	Tonopah	20 Flight Lines	1174
	Cessna-206	Fallon	24 Partial Flight Lines, 1 Tie Line	1010
10.03.22	Bell Jet Ranger	Tonopah	6 Flight Lines	387
	Cessna-206	Fallon	6 Partial Flight Lines, 1 Tie Line	1086
10.04.22	Bell Jet Ranger	Tonopah	26 Flight Lines	1650
	Cessna-206	Fallon	12 Flight Lines	1032
10.05.22	Bell Jet Ranger	Tonopah	22 Flight Lines	1492
	Cessna-206	Fallon	8 Flight Lines, 2 Tie Lines	766
10.06.22	Bell Jet Ranger	Tonopah	36 Flight Lines, 4 Tie Lines (Completes Tonopah Block)	1225
	Cessna-206	Fallon	22 Partial Flight Lines	570
10.07.22	Cessna-206	Fallon	9 Flight Lines, 2 tie Lines	288
10.08.22	Cessna-206	Fallon	25 Partial Flight Lines	1305

<u>Date</u>	<u>Aircraft</u>	<u>Base</u>	<u>Comments</u>	<u>Kms</u>
10.09.22	Cessna-206	Fallon	12 Partial Flight Lines, 10 Tie Lines	1124
10.10.22	Cessna-206	Fallon	16 Partial Flight Lines	1000
10.11.22	Cessna-206	Fallon	40 Partial Flight Lines	1100
10.12.22	Cessna-206	Fallon	26 Partial Flight Lines	906
10.13.22	Cessna-206	Fallon	22 Partial Flight Lines	1271
10.14.22	Cessna-206	Fallon	20 Partial Flight Lines	1009
10.15.22	Cessna-206	Fallon	19 Partial Flight Lines	797
10.16.22	Cessna-206	Fallon	32 Partial Flight Lines	1218
10.17.22	Cessna-206	Fallon	16 Partial Flight Lines	1098
10.18.22	Cessna-206	Fallon	8 Flight Lines	736
10.19.22	Cessna-206	Fallon	10 Flight Lines, 16 Partial Flight Lines	1137
10.20.22	Cessna-206	Fallon	12 Flight Lines, 1 Partial Flight Line	592
10.21.22	Cessna-206	Fallon	17 Flight Lines, 1 Tie Line	1416
10.23.22	Cessna-206	Fallon	47 Partial Flight Lines	1192
10.24.22	Cessna-206	Fallon	16 Flight Lines, 2 Partial Tie Lines	1358
10.25.22	Cessna-206	Fallon	12 Flight Lines, 10 Partial Flight Lines	1488
10.26.22	Cessna-206	Fallon	16 Flight Lines, 2 Partial Tie Lines	1395
10.27.22	Cessna-206	Fallon	6 Flight Lines, 2 Partial Tie Lines	613
10.28.22	Cessna-206	Fallon	10 Flight Lines, 2 Partial Tie Lines	1139
10.29.22	Cessna-206	Fallon	30 Partial Flight Lines	1520
10.30.22	Cessna-206	Fallon	20 Partial Flight Lines	788
	C-206 Crew Repositioned to Hawthorne			
10.31.22	Cessna-206	Hawthorne	4 Flight Lines, 2 Partial Flight Lines	495
11.02.22	Cessna-206	Hawthorne	2 Fight Lines, 4 Partial Flight Lines	404
11.04.22	Cessna-206	Hawthorne	10 Flight Lines, 2 Partial Flight Lines, 4 Tie Lines	916
11.05.22	Cessna-206	Hawthorne	2 Flight Lines, 11 Partial Flight Lines	728
11.06.22	Cessna-206	Hawthorne	2 Flight Lines, 6 Partial Flight Lines	428
11.09.22	Cessna-206	Hawthorne	4 Flight Lines	376
11.10.22	Cessna-206	Hawthorne	10 Flight Lines, 6 Partial Flight Lines, 2 Tie Lines	1124
11.11.22	Cessna-206	Hawthorne	18 Flight Lines, 2 Tie Lines	1021
11.12.22	Cessna-206	Hawthorne	8 Flight Lines, 4 Tie Lines	865
11.14.22	Cessna-206	Hawthorne	13 Flight Lines, 3 Tie Lines	1285
11.15.22	Cessna-206	Hawthorne	14 Flight Lines	1274
11.16.22	Cessna-206	Hawthorne	20 Flight Lines, 2 Tie Lines	1161
11.17.22	Cessna-206	Hawthorne	9 Flight Lines, 5 Tie Lines	903
11.19.22	Cessna-206	Hawthorne	20 Flight Lines, 2 Tie Lines	1210
11.20.22	Cessna-206	Hawthorne	18 Flight Lines	886

Appendix B: Control Line Mean Counts

Fixed-Wing (Winnemucca, Fallon and Hawthorne Blocks)

Date	Mean Total Counts/Sec	Mean Th Counts/Sec	Mean U Counts/Sec	Mean K Counts/Sec	Mean Humidity %	Mean Temp °C	Flight #	Location
10/30/2021	2109	86	84	108	58	19.7	1	Winnemucca
10/31/2021	1671	69	68	87	55	16.7	2	Winnemucca
11/1/2021	1824	75	70	91	58	12.0	3,4	Winnemucca
11/2/2021	1973	83	79	101	58	9.2	5,6	Winnemucca
11/3/2021	2087	87	85	106	61	9.8	7	Winnemucca
11/5/2021	1924	81	78	100	57	11.2	8	Winnemucca
11/7/2021	1674	71	70	87	54	6.5	9	Winnemucca
11/8/2021	2080	88	83	106	58	3.0	10	Winnemucca
11/11/2021	1672	69	70	88	64	16.2	11	Winnemucca
11/12/2021	1859	77	76	97	76	11.7	12,13	Winnemucca
11/13/2021	2049	83	81	104	55	11.1	14,15	Winnemucca
11/14/2021	1951	80	78	100	54	12.5	16,17	Winnemucca
11/17/2021	1885	78	78	98	53	5.6	18	Winnemucca
11/20/2021	1899	77	76	96	53	3.7	19,20	Winnemucca
11/21/2021	1939	80	78	100	55	9.2	21,22	Winnemucca
11/22/2021	2111	86	81	104	53	4.7	23	Winnemucca
11/25/2021	1847	79	75	96	53	1.7	24,25	Winnemucca
11/26/2021	2194	96	86	110	54	5.2	26,27	Winnemucca
11/27/2021	2294	95	88	113	54	6.3	28,29	Winnemucca
11/28/2021	2263	95	87	112	54	8.9	30,31	Winnemucca
11/30/2021	2203	92	85	110	53	7.4	32,33	Winnemucca
12/1/2021	2273	94	88	112	53	7.7	34	Winnemucca
12/2/2021	2395	97	91	117	53	4.8	35,36	Winnemucca
12/3/2021	2441	100	94	121	52	13.7	37,38	Winnemucca

Date	Mean Total Counts/Sec	Mean Th Counts/Sec	Mean U Counts/Sec	Mean K Counts/Sec	Mean Humidity %	Mean Temp °C	Flight #	Location
12/4/2021	2483	101	92	120	55	7.6	39	Winnemucca
12/7/2021	2375	96	89	115	52	4.6	40,41	Winnemucca
5/15/2022	1797	75	72	91	43	16.3	43	Winnemucca
5/16/2022	1998	82	83	104	45	14.	44,45	Winnemucca
5/17/2022	2030	83	82	103	47	11.0	46,47,48,49	Winnemucca
5/18/2021	2011	82	81	103	31	14.3	50	Winnemucca
5/21/2022	1912	79	78	98	31	9.8	51	Winnemucca
5/22/2022	1790	75	72	92	33	6.8	52	Winnemucca
5/23/2022	1875	77	77	97	31	11.4	53	Winnemucca
5/24/2022	1742	71	73	92	19	22.8	54	Winnemucca
5/25/2022	2547	102	105	129	35	10.0	55	Winnemucca
5/26/2022	2111	87	87	108	33	13.9	56	Winnemucca
5/27/2022	2163	90	89	113	38	11.4	57	Winnemucca
5/31/2022	1826	77	107	96	57	7.0	58	Winnemucca
6/1/2022	2613	107	108	131	47	7.3	59	Winnemucca
6/2/2022	2128	88	86	110	33	10.7	60	Winnemucca
6/3/2022	2528	103	107	129	40	14.2	61	Winnemucca
6/6/2022	1738	73	73	92	50	12.6	62	Winnemucca
6/7/2022	2510	102	106	127	48	10.3	63	Winnemucca
6/8/2022	2045	85	83	104	34	13.2	64	Winnemucca
6/9/2022	2604	104	107	132	45	16.2	65	Winnemucca
6/10/2022	2058	85	85	106	41	17.3	66	Winnemucca
6/14/2022	2627	108	109	132	44	5.3	67	Winnemucca
6/15/2022	1955	80	80	101	37	9.3	68	Winnemucca
6/16/2022	2617	106	107	133	28	12.4	69	Winnemucca
6/20/2022	1986	83	82	104	47	7.5	70	Winnemucca
6/22/2022	2557	102	107	130	34	14.6	71	Winnemucca
6/24/2022	2497	101	103	127	41	14.6	72	Winnemucca
6/26/2022	2567	103	107	131	26	18.2	73	Winnemucca

Date	Mean Total Counts/Sec	Mean Th Counts/Sec	Mean U Counts/Sec	Mean K Counts/Sec	Mean Humidity %	Mean Temp °C	Flight #	Location
6/27/2022	2820	113	116	141	32	16.4	74	Winnemucca
6/29/2022	2074	85	86	106	26	16.1	75	Winnemucca
6/30/2022	2579	105	107	130	28	15.1	76	Winnemucca
7/1/2022	2587	104	108	132	21	14.5	77	Winnemucca
7/2/2022	1800	74	76	0	12	27.8	78	Winnemucca
7/3/2022	2033	86	85	108	25	16.2	79	Winnemucca
7/4/2022	1912	80	79	101	28	13.7	80	Winnemucca
7/5/2022	1972	82	82	103	35	16.5	81	Winnemucca
7/6/2022	2412	97	102	124	39	17.1	82	Winnemucca
7/7/2022	2527	101	107	128	36	17.3	83	Winnemucca
7/8/2022	2568	102	108	130	28	17.1	84	Winnemucca
7/9/2022	1976	82	83	104	23	17.9	85	Winnemucca
7/10/2022	2592	105	107	130	27	14.4	86	Winnemucca
7/11/2022	2292	93	97	118	30	18.3	87	Winnemucca
7/14/2022	1733	74	74	91	28	18.9	88	Fallon
7/15/2022	1981	86	84	104	25	18.7	89	Fallon
7/16/2022	1798	77	76	94	42	20.9	90	Fallon
7/17/2022	1932	83	84	103	29	21.0	91	Fallon
7/20/2022	1835	79	77	96	46	19.3	92	Fallon
7/21/2022	1725	74	73	91	27	20.0	93	Fallon
7/23/2022	1640	71	70	87	21	18.6	94	Fallon
7/24/2022	1620	70	68	85	22	18.6	95	Fallon
7/31/2022	1958	85	81	102	25	22.1	96	Fallon
8/1/2022	1712	73	73	91	46	21.6	97	Fallon
8/2/2022	1792	77	75	94	46	20.1	98	Fallon
8/3/2022	1719	74	73	91	50	20.4	99	Fallon
8/4/2022	1672	72	69	88	46	20.5	100	Fallon
8/5/2022	1716	74	71	90	50	22.0	101	Fallon
8/6/2022	1611	69	68	84	75	18.0	102	Fallon

Date	Mean Total Counts/Sec	Mean Th Counts/Sec	Mean U Counts/Sec	Mean K Counts/Sec	Mean Humidity %	Mean Temp °C	Flight #	Location
8/7/2022	1885	81	81	98	53	19.5	103	Fallon
8/8/2022	1702	73	72	90	37	22.5	104	Fallon
8/9/2022	1598	69	66	83	65	18.8	105	Fallon
8/10/2022	1847	80	78	98	76	17.3	106	Fallon
8/11/2022	1712	75	72	90	51	19.0	107	Fallon
8/12/2022	1827	79	77	98	31	18.6	108	Fallon
8/13/2022	1584	69	68	84	24	17.5	109	Fallon
8/14/2022	1854	81	79	99	28	17.8	110	Fallon
8/17/2022	1934	83	80	100	36	22.2	111	Fallon
9/26/2022	1545	67	65	82	17	29.5	113	Fallon
9/27/2022	1674	71	68	86	34	19.6	114	Fallon
9/28/2022	1797	77	76	94	35	15.9	115	Fallon
9/29/2022	1827	80	78	97	37	15.0	116	Fallon
9/30/2022	1802	78	75	94	41	13.6	117	Fallon
10/1/2022	1923	81	79	100	49	11.4	118	Fallon
10/2/2022	1852	80	75	95	49	10.6	119	Fallon
10/3/2022	2028	86	82	104	49	10.2	120	Fallon
10/4/2022	1919	82	78	98	43	12.1	121	Fallon
10/5/2022	1988	86	79	102	41	13.3	122	Fallon
10/6/2022	2089	89	84	107	38	13.5	123	Fallon
10/7/2022	2113	88	84	109	29	17.5	124	Fallon
10/8/2022	1993	86	80	102	32	13.3	125	Fallon
10/9/2022	2039	87	82	105	36	12.6	126	Fallon
10/10/2022	1975	86	81	103	39	14.3	127	Fallon
10/11/2022	2059	87	83	106	43	12.8	128	Fallon
10/12/2022	1868	80	77	97	37	12.4	129	Fallon
10/13/2022	2056	87	84	105	42	10.9	130	Fallon
10/14/2022	2006	86	82	103	38	11.1	131	Fallon
10/15/2022	1891	82	76	96	37	11.2	132	Fallon

Date	Mean Total Counts/Sec	Mean Th Counts/Sec	Mean U Counts/Sec	Mean K Counts/Sec	Mean Humidity %	Mean Temp °C	Flight #	Location
10/16/2022	1891	81	77	97	34	11.0	133	Fallon
10/17/2022	2101	91	84	107	39	8.7	134	Fallon
10/18/2022	2122	89	83	107	25	17.5	135	Fallon
10/19/2022	2061	87	82	104	33	10.5	136	Fallon
10/20/2022	1976	84	80	101	31	9.7	137	Fallon
10/21/2022	2073	87	80	104	36	9.1	138	Fallon
10/23/2022	1703	73	72	90	51	6.9	139	Fallon
10/24/2022	1595	69	67	84	49	3.2	140	Fallon
10/25/2022	1853	81	75	95	47	9.4	141	Fallon
10/26/2022	1688	74	71	88	54	7.1	142	Fallon
10/27/2022	1814	76	74	94	23	14.1	143	Fallon
10/28/2022	1877	81	75	97	37	7.5	144	Fallon
10/29/2022	1984	85	79	102	39	5.2	145	Fallon
10/30/2022	1981	86	79	101	44	4.9	146	Fallon
10/31/2022	3007	121	119	151	24	17.7	147	Hawthorne
11/2/2022	2318	97	96	122	35	7.6	148	Hawthorne
11/4/2022	2294	95	95	119	47	5.6	149	Hawthorne
11/5/2022	2784	114	113	142	53	8.3	150	Hawthorne
11/6/2022	2491	102	103	130	62	11.9	151	Hawthorne
11/9/2022	1767	74	75	93	45	9.9	152	Hawthorne
11/10/2022	2257	93	92	117	55	4.8	153	Hawthorne
11/11/2022	2460	101	99	125	60	4.1	154	Hawthorne
11/12/2022	2301	94	91	118	44	7.5	155	Hawthorne
11/14/2022	2768	112	111	140	67	4.2	156	Hawthorne
11/15/2022	2647	108	104	133	66	5.3	157	Hawthorne
11/16/2022	2459	100	97	124	59	5.3	158	Hawthorne
11/17/2022	2539	103	99	127	56	4.8	159	Hawthorne
11/19/2022	2538	104	99	128	56	2.3	160	Hawthorne
11/20/2022	2256	93	88	114	23	11.1	161	Hawthorne

Bell Jet Ranger (Tonopah Block)

Date	Mean Total Counts/Sec	Mean Th Counts/Sec	Mean U Counts/Sec	Mean K Counts/Sec	Mean Humidity %	Mean Temp °C	Flight #	Location
9/12/2022	3197	133	142	173	32	27.8	1	Tonopah
9/13/2022	3629	150	162	192	34	27.2	2	Tonopah
9/14/2022	3866	159	169	207	54	20.0	3	Tonopah
9/15/2022	3417	143	149	183	22	28.2	4	Tonopah
9/16/2022	3729	154	163	200	26	23.1	5	Tonopah
9/17/2022	2750	116	120	149	13	29.3	6	Tonopah
9/19/2022	2662	113	119	146	42	11.2	7	Tonopah
9/20/2022	3000	125	132	161	36	10.2	8	Tonopah
9/21/2022	3166	135	142	169	39	14.8	9	Tonopah
9/22/2022	2954	124	128	159	49	10.2	10	Tonopah
9/23/2022	3924	162	170	207	36	10.4	11	Tonopah
9/26/2022	3364	140	145	179	19	13.3	12	Tonopah
9/27/2022	2509	104	109	137	8	28.5	13	Tonopah
9/28/2022	3764	154	167	200	28	13.8	14	Tonopah
9/29/2022	3752	154	165	196	53	14.4	15	Tonopah
9/30/2022	4011	166	175	213	31	12.3	16	Tonopah
10/1/2022	4035	167	176	211	31	12.5	17	Tonopah
10/2/2022	4080	168	177	213	31	13.0	18	Tonopah
10/3/2022	2815	120	121	150	13	25.8	19	Tonopah
10/4/2022	3362	141	143	180	32	11.1	20	Tonopah
10/5/2022	3645	152	158	195	26	12.6	21	Tonopah
10/6/2022	4134	170	180	216	24	14.2	22	Tonopah

Appendix C: Daily Spectrometer Resolution Tests

Fixed-Wing (Winnemucca, Fallon and Hawthorne Blocks)

Date	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Summed	Summed	Summed	Summed
	110095	110095	110096	110096	110097	110097	110098	110098	180510	180510	Down	Down	Up	Up
	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137
11/1/2021	4.24	7.66	5.46	8.80	5.07	9.01	5.21	8.44	3.44	8.28	4.97	8.51	3.44	8.28
11/2/2021	4.76	7.17	5.40	7.96	5.08	8.28	5.36	8.34	4.56	4.72	5.30	7.70	4.56	4.72
11/3/2021	4.52	7.22	5.59	9.05	4.89	8.43	5.37	8.33	4.38	6.36	5.09	8.96	4.38	6.36
11/5/2021	4.73	6.36	5.53	8.32	4.77	8.41	5.14	8.68	4.12	5.74	5.10	7.88	4.12	5.74
11/7/2021	4.79	7.97	5.61	8.44	5.07	6.81	6.05	6.88	3.55	7.56	5.65	7.15	3.55	7.56
11/8/2021	4.44	7.38	5.61	9.10	5.03	8.72	4.92	8.38	4.70	8.15	5.34	9.10	4.70	8.15
11/11/2021	4.95	6.15	5.37	4.99	4.87	9.41	5.55	7.91	4.56	7.15	5.26	8.15	4.56	7.15
11/12/2021	4.53	7.98	5.53	8.04	4.64	8.78	5.30	8.54	4.27	5.49	5.08	7.98	4.27	5.49
11/13/2021	4.60	8.09	5.13	8.90	4.67	8.34	5.25	8.18	4.60	5.62	4.92	7.96	4.60	5.62
11/14/2021	4.76	7.38	5.31	9.07	4.91	8.59	8.99	4.08	6.52	4.76	5.29	8.64	4.08	6.52
11/17/2021	3.88	7.73	4.52	7.73	4.86	8.72	5.42	8.14	5.54	7.16	5.38	8.51	5.54	7.16
11/20/2021	4.67	8.15	5.50	8.09	4.91	8.39	6.96	8.90	4.94	5.41	5.20	7.92	4.94	5.41
11/21/2021	4.74	8.03	5.28	7.96	4.63	7.91	6.28	8.16	4.19	4.52	5.14	8.20	4.19	4.52
11/22/2021	4.71	7.44	5.79	9.03	4.75	8.77	5.17	9.45	3.73	6.99	5.13	8.87	3.73	6.99
11/25/2021	4.24	6.30	5.10	8.46	4.83	8.02	6.20	9.57	3.24	7.83	5.24	8.75	3.24	7.83
11/26/2021	4.75	7.59	5.56	8.83	4.87	8.79	5.91	9.78	2.75	8.45	5.33	8.58	2.75	8.45
11/27/2021	4.50	7.82	4.37	9.46	4.95	8.98	5.00	9.36	3.31	7.54	5.23	8.77	3.31	7.54
11/28/2021	4.58	7.74	5.70	9.50	5.05	9.21	4.92	9.22	3.84	7.27	5.02	8.94	3.84	7.27
11/30/2021	4.63	8.11	5.98	10.17	4.81	9.33	5.45	10.45	4.46	8.17	4.77	9.14	4.46	8.17
12/1/2021	4.42	7.81	5.36	9.76	4.80	8.11	6.29	10.37	4.00	8.11	5.11	8.99	4.00	8.11
12/2/2021	4.81	7.79	5.60	9.43	4.72	8.99	3.98	7.03	3.96	7.21	4.65	8.08	3.96	7.21
12/3/2021	4.92	8.01	5.18	9.73	5.01	9.31	3.94	7.05	3.79	7.61	4.71	8.24	3.79	7.61
12/4/2021	4.89	7.77	5.53	9.16	4.62	9.25	3.90	7.00	3.65	8.10	4.67	8.10	3.65	8.10

Date	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Summed	Summed	Summed	Summed
	110095	110095	110096	110096	110097	110097	110098	110098	180510	180510	Down	Down	Up	Up
	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137
12/7/2021	4.73	7.70	5.87	8.96	4.95	8.52	3.76	6.51	3.83	6.67	4.77	7.53	3.83	6.67
5/15/2022	4.73	6.08	5.50	9.06	4.93	9.23	3.72	8.15	4.85	5.54	4.68	8.53	4.85	5.54
5/16/2022	4.54	7.80	5.30	6.04	4.90	6.50	4.11	5.38	4.80	4.69	4.67	5.59	4.80	4.69
5/17/2022	4.56	6.38	4.91	8.13	4.55	8.19	3.71	5.82	3.98	6.12	4.60	8.37	3.98	6.12
5/18/2022	4.52	7.74	5.81	7.86	4.93	7.80	4.03	8.11	4.14	3.98	4.62	8.46	4.14	3.98
5/21/2022	4.48	7.99	5.71	7.68	4.98	6.17	3.90	8.30	4.61	4.30	4.70	5.87	4.61	4.30
5/22/2022	4.53	6.92	5.61	7.58	4.94	9.20	4.09	7.94	4.10	5.33	4.65	8.44	4.10	5.33
5/23/2022	4.56	8.27	5.53	7.45	4.77	5.74	3.91	5.35	4.18	3.45	4.66	5.65	4.18	3.45
5/24/2022	4.62	4.18	5.00	7.38	4.64	3.62	4.09	7.93	4.27	4.34	4.48	5.61	4.27	4.34
5/25/2022	4.58	7.98	5.61	6.07	4.92	7.28	4.02	8.09	3.57	4.14	4.63	8.39	3.57	4.14
5/26/2022	4.37	5.79	4.12	8.47	5.01	8.00	3.76	5.84	4.63	5.82	4.58	6.40	4.63	5.82
5/27/2022	4.25	5.74	5.48	7.73	4.81	8.70	3.91	4.95	4.32	4.13	4.55	5.63	4.32	4.13
5/31/2022	4.72	7.83	5.75	6.90	4.85	6.61	4.04	8.06	4.18	1.99	4.73	5.77	4.18	1.99
6/1/2022	4.60	8.44	5.42	7.46	5.01	7.10	3.97	8.01	3.95	4.87	4.66	8.40	3.95	4.87
6/2/2022	4.58	6.10	5.24	7.85	4.87	8.76	4.06	5.75	4.24	5.71	4.65	8.22	4.24	5.71
6/3/2022	4.60	8.08	5.47	6.53	4.88	6.56	3.91	7.96	4.39	2.95	4.67	5.82	4.39	2.85
6/6/2022	4.57	4.46	5.71	6.30	4.88	7.86	4.06	7.92	4.20	4.98	4.64	5.26	4.20	4.98
6/7/2022	4.63	6.38	5.67	7.88	4.81	7.63	4.05	8.00	3.86	4.57	4.63	6.58	3.86	4.57
6/8/2022	4.54	8.12	5.47	6.69	4.84	7.32	3.95	7.97	3.71	4.43	4.64	8.38	3.71	4.43
6/9/2022	4.37	7.04	5.49	7.74	5.02	6.77	3.92	5.44	4.08	5.09	4.61	8.35	4.08	5.09
6/10/2022	4.36	7.77	5.05	7.94	4.91	8.65	3.91	7.89	3.61	4.56	4.63	8.28	3.61	4.56
6/14/2022	4.74	7.51	5.48	7.24	4.98	6.62	3.87	7.84	4.27	4.32	4.63	4.82	4.27	4.32
6/15/2022	4.65	7.81	5.56	8.45	4.87	7.21	4.07	7.79	4.38	4.89	4.61	6.13	4.38	4.89
6/16/2022	4.42	5.69	5.57	8.26	5.09	8.83	4.00	6.02	4.79	5.96	4.59	8.29	4.79	5.96
6/20/2022	4.76	5.56	5.70	7.62	4.94	7.42	3.99	7.93	4.40	5.96	4.72	5.84	4.40	5.96
6/22/2022	4.63	8.43	5.65	7.79	4.95	8.49	4.05	8.16	4.01	4.61	4.70	8.41	4.01	4.61
6/24/2022	4.54	8.05	5.18	6.36	4.75	7.40	3.81	7.92	4.48	4.88	4.53	8.50	4.48	4.88
6/26/2022	4.57	7.64	5.43	7.24	4.86	7.79	4.03	5.69	3.78	5.03	4.69	6.36	3.78	5.03

Date	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Summed	Summed	Summed	Summed
	110095	110095	110096	110096	110097	110097	110098	110098	180510	180510	Down	Down	Up	Up
	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137
6/27/2022	4.40	8.04	5.35	7.30	4.44	7.02	4.08	5.16	4.00	3.98	4.52	5.86	4.00	3.98
6/29/2022	4.34	7.37	4.34	7.54	4.94	7.75	3.92	2.85	4.15	6.99	4.54	3.83	4.15	6.99
6/30/2022	4.44	4.28	5.53	8.34	4.62	6.67	3.99	3.52	4.55	5.28	4.49	3.81	4.55	5.28
7/1/2022	4.28	8.25	5.53	8.35	4.90	6.66	3.88	8.04	3.22	4.27	4.56	5.67	4.27	4.56
7/2/2022	4.51	8.11	4.78	3.79	4.79	5.85	3.45	7.84	4.69	6.90	4.62	7.10	4.69	6.90
7/3/2022	4.22	8.04	5.46	7.76	4.67	8.43	3.90	4.95	4.51	5.29	4.60	5.12	4.51	5.29
7/4/2022	4.74	6.10	5.29	7.52	4.74	8.23	3.97	4.19	4.31	8.23	4.56	5.88	4.31	8.23
7/5/2022	4.38	3.75	5.44	7.70	5.06	6.24	4.00	7.96	4.15	7.21	4.61	7.73	4.15	7.21
7/6/2022	4.53	3.41	5.31	5.73	4.68	6.16	3.96	4.83	3.80	7.80	4.68	6.15	3.80	7.80
7/7/2022	4.64	7.33	5.41	5.60	4.47	7.24	3.89	3.31	4.13	7.09	4.39	6.97	4.13	7.09
7/8/2022	4.05	4.20	5.28	8.62	4.80	5.85	3.89	4.43	3.54	6.96	4.56	5.42	3.54	6.96
7/9/2022	4.43	3.18	4.73	7.54	4.86	6.70	3.98	3.30	4.00	7.36	4.64	7.29	4.00	7.36
7/10/2022	4.39	5.53	5.66	6.45	4.77	8.00	4.02	5.42	3.98	8.04	4.61	5.92	3.98	8.04
7/11/2022	4.33	5.58	5.43	7.01	4.82	7.56	4.06	5.18	3.83	4.12	4.64	5.96	3.83	4.12
7/14/2022	4.39	5.33	5.19	6.94	4.35	5.67	3.99	5.11	3.93	2.80	4.60	5.68	3.93	2.80
7/15/2022	4.37	5.67	5.31	8.13	4.70	6.86	4.02	8.16	3.94	4.01	4.58	6.16	3.94	4.01
7/16/2022	4.23	8.16	5.32	2.34	4.72	7.43	3.92	8.11	3.87	5.07	4.53	6.41	3.87	5.07
7/17/2022	4.22	7.55	5.08	8.27	4.69	7.21	4.04	7.87	4.28	2.98	4.54	5.82	4.28	2.98
7/20/2022	4.54	8.53	5.20	7.73	4.93	7.09	3.97	5.31	4.69	3.53	4.65	5.90	4.69	3.53
7/21/2022	4.31	8.00	5.49	6.00	4.77	5.63	4.10	4.40	3.98	4.03	4.53	4.91	3.98	4.03
7/23/2022	4.42	7.99	5.34	8.84	4.77	7.05	3.95	5.60	3.67	4.61	4.56	6.16	3.67	4.61
7/24/2022	4.34	6.79	5.10	8.66	4.53	8.01	3.99	6.35	4.13	6.28	4.47	7.19	4.13	6.28
7/31/2022	4.44	6.23	5.09	8.91	5.05	8.77	4.12	5.96	3.73	6.00	4.61	7.03	3.73	6.00
8/1/2022	4.47	7.79	5.07	7.63	4.74	7.42	4.06	8.00	4.06	5.38	4.58	8.35	4.06	5.38
8/2/2022	4.54	5.88	5.39	7.19	4.84	8.00	4.01	8.06	3.86	3.39	4.53	6.22	3.86	3.39
8/3/2022	4.31	6.48	5.76	8.36	4.60	7.43	4.01	7.79	4.16	4.96	4.60	8.31	4.16	4.96
8/4/2022	4.59	4.49	4.83	6.85	4.78	6.21	4.01	7.76	4.20	4.81	4.60	8.10	4.20	4.81
8/5/2022	4.25	6.44	4.55	7.97	4.75	8.55	3.98	6.16	4.13	6.29	4.64	8.07	4.13	6.29

Date	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Summed	Summed	Summed	Summed
	110095	110095	110096	110096	110097	110097	110098	110098	180510	180510	Down	Down	Up	Up
	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137
8/6/2022	4.55	8.70	5.57	8.53	4.82	6.25	3.98	8.01	3.25	3.39	4.72	7.93	3.25	3.39
8/7/2022	4.55	8.33	5.34	7.98	5.10	7.31	4.01	8.08	3.96	5.07	4.59	8.35	3.96	5.07
8/8/2022	4.12	4.92	5.17	8.68	4.74	6.61	3.83	7.99	3.73	4.70	4.58	8.30	3.73	4.70
8/9/2022	4.41	6.45	5.52	8.12	4.96	8.81	3.95	8.03	4.15	5.61	4.57	8.49	4.15	5.61
8/10/2022	4.38	5.78	5.03	7.88	4.76	6.92	3.90	8.14	4.25	4.24	4.50	6.45	4.25	4.24
8/11/2022	4.38	5.74	5.54	7.24	4.76	8.87	4.00	8.09	3.87	5.34	4.60	8.43	3.87	5.34
8/12/2022	4.48	8.47	5.39	8.78	4.58	6.45	4.00	7.72	3.42	2.86	4.55	5.57	3.42	2.86
8/13/2022	4.34	3.88	5.28	8.52	4.78	7.35	3.88	8.20	4.09	4.02	4.60	4.36	4.09	4.06
8/14/2022	4.52	6.21	5.26	7.54	4.87	7.04	3.86	7.93	3.60	4.07	4.56	6.21	4.07	3.86
8/17/2022	4.29	6.84	5.40	8.63	4.69	7.61	3.86	6.44	3.61	6.64	4.57	7.05	3.61	6.64
9/26/2022	4.54	8.10	5.23	7.81	4.97	7.27	3.88	8.20	4.15	4.76	4.74	6.63	4.15	4.76
9/27/2022	4.57	6.55	5.47	8.51	4.55	8.78	3.95	6.05	4.23	5.72	4.66	8.45	4.23	5.72
9/28/2022	4.59	8.28	5.38	8.85	3.94	8.26	4.76	8.38	4.34	6.64	4.56	8.67	4.34	6.64
9/29/2022	4.50	8.26	5.40	7.97	4.00	8.17	5.12	7.27	4.07	5.27	4.60	6.73	4.07	5.27
9/30/2022	4.60	8.37	5.38	7.09	4.90	9.13	3.95	8.22	4.02	4.88	4.69	8.55	4.02	4.88
10/1/2022	4.71	6.92	5.48	8.93	4.96	7.95	3.90	5.87	4.05	6.28	4.71	8.27	4.05	6.28
10/2/2022	4.54	7.30	5.47	8.88	4.97	8.57	3.79	6.47	4.10	6.79	4.60	7.67	4.10	6.79
10/3/2022	4.59	7.18	5.56	8.88	4.80	8.33	3.86	6.38	4.00	6.60	4.72	7.36	4.00	6.60
10/4/2022	4.50	6.96	5.39	8.85	4.94	7.92	3.92	6.32	3.93	6.66	4.65	7.34	3.93	6.66
10/5/2022	4.19	6.92	5.68	8.64	4.80	8.32	3.90	6.64	4.45	6.59	4.64	7.30	4.45	6.59
10/6/2022	4.62	6.64	5.62	8.41	4.44	7.96	4.01	6.32	3.30	6.28	4.60	6.92	3.30	6.28
10/7/2022	4.43	7.23	5.37	8.73	4.88	8.14	3.86	6.49	3.88	6.86	4.52	7.34	3.88	6.86
10/8/2022	4.59	7.66	5.38	9.31	4.81	8.65	3.83	6.98	4.10	7.40	4.52	7.84	4.09	7.39
10/9/2022	4.70	7.26	5.48	9.32	5.00	8.39	4.01	6.74	4.02	6.84	4.61	7.56	4.02	6.84
10/10/2022	4.70	6.63	5.61	8.94	4.89	9.22	3.93	5.95	4.19	6.30	4.58	8.50	4.19	6.30
10/11/2022	4.21	4.40	5.53	2.08	5.11	6.81	4.07	4.68	4.23	4.63	4.64	8.32	4.23	4.63
10/12/2022	4.36	5.56	5.37	8.30	5.08	8.91	4.10	5.46	3.75	4.99	4.64	8.41	3.75	4.99
10/13/2022	4.47	6.87	5.66	9.05	4.94	8.17	4.05	6.36	4.27	6.57	4.71	7.17	4.27	6.57

Date	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Summed	Summed	Summed	Summed
	110095	110095	110096	110096	110097	110097	110098	110098	180510	180510	Down	Down	Up	Up
	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137
10/14/2022	4.54	7.11	5.29	8.96	4.75	8.45	3.93	6.82	3.70	7.31	4.56	7.59	3.70	7.31
10/15/2022	4.48	6.91	5.56	8.64	4.58	8.43	3.82	6.25	3.91	6.67	4.57	7.20	3.91	6.67
10/16/2022	4.61	7.12	5.50	8.97	5.04	8.27	3.85	6.57	4.16	6.71	4.58	7.27	4.16	6.71
10/17/2022	4.57	7.69	5.53	9.36	4.91	8.68	3.81	6.82	3.71	7.31	4.58	7.81	3.71	7.31
10/18/2022	4.29	7.51	5.14	9.27	4.99	8.60	3.84	6.76	3.70	8.17	4.55	7.73	3.70	8.17
10/19/2022	4.66	7.55	5.46	9.05	5.20	8.65	3.89	6.77	4.19	7.10	4.61	7.81	4.19	7.10
10/20/2022	4.49	7.59	5.57	9.42	4.94	8.68	3.87	6.70	3.83	7.26	4.62	7.82	3.83	7.26
10/21/2022	4.58	7.74	5.80	9.11	4.94	8.69	4.01	7.02	3.97	7.34	4.70	7.92	3.97	7.34
10/23/2022	4.75	8.45	5.97	7.59	4.89	7.49	4.06	7.95	4.18	5.80	4.66	8.45	4.18	5.80
10/24/2022	4.75	6.69	6.12	8.91	5.06	8.92	4.03	6.25	4.52	6.59	4.80	7.15	4.52	6.59
10/25/2022	4.62	6.23	5.83	8.27	5.10	8.93	4.02	8.06	4.18	6.06	4.75	8.44	4.18	6.06
10/26/2022	4.77	8.08	6.06	8.08	4.97	8.72	4.05	8.15	4.58	5.45	4.73	8.28	4.58	5.45
10/27/2022	4.94	6.11	5.43	7.84	4.83	7.84	3.86	5.56	4.82	5.98	4.70	8.52	4.82	5.98
10/28/2022	4.65	6.66	5.60	9.16	4.95	8.95	3.92	6.45	3.74	6.06	4.68	7.37	3.74	6.06
10/29/2022	4.89	7.56	5.85	9.42	4.78	8.52	3.85	6.81	4.35	7.23	4.65	7.75	4.35	7.23
10/30/2022	4.79	7.54	5.79	9.07	4.98	8.34	3.96	6.70	4.53	7.14	4.73	7.47	4.53	7.14
10/31/2022	4.37	7.13	5.58	9.30	4.73	8.50	3.78	6.66	3.73	7.74	4.46	7.64	3.73	7.74
11/2/2022	4.65	6.81	5.95	8.07	4.75	8.58	3.80	7.98	4.50	6.12	4.84	8.51	4.50	6.12
11/4/2022	4.95	7.71	5.69	8.10	4.96	8.02	3.99	8.05	4.50	4.90	4.74	6.16	4.50	4.90
11/5/2022	4.81	6.88	5.64	8.82	4.39	8.79	3.97	6.08	3.26	6.55	4.68	7.15	3.26	6.55
11/6/2022	4.60	6.24	5.51	9.27	5.08	7.34	3.87	5.83	5.06	5.95	4.63	6.70	5.06	5.95
11/9/2022	4.60	7.93	5.62	8.-4	4.94	7.96	3.76	7.80	4.43	4.79	4.57	7.79	4.43	4.79
11/10/2022	4.90	6.07	5.83	7.76	4.67	7.10	3.93	8.22	4.13	4.33	4.73	8.29	4.13	4.33
11/11/2022	4.89	6.26	5.42	8.22	4.87	7.44	3.78	5.97	4.02	6.29	4.76	6.65	4.02	6.29
11/12/2022	4.95	7.35	5.91	9.23	4.95	8.28	3.94	6.44	4.53	7.17	4.76	7.56	4.53	7.17
11/14/2022	4.74	6.94	5.76	8.85	4.84	8.30	3.88	6.61	3.72	6.67	4.64	7.21	3.72	6.67
11/15/2022	4.68	7.24	5.71	9.12	4.96	8.27	3.93	6.48	3.31	6.71	4.72	7.42	3.31	6.71
11/16/2022	4.71	7.34	5.95	9.27	5.12	8.26	4.12	6.60	3.31	6.73	4.76	7.60	3.31	6.73

Date	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Summed	Summed	Summed	Summed
	110095	110095	110096	110096	110097	110097	110098	110098	180510	180510	Down	Down	Up	Up
	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137	Tl208	Cs137
11/17/2022	4.89	7.35	5.82	8.69	4.84	8.32	4.02	6.64	4.24	6.72	4.80	7.41	4.24	6.72
11/19/2022	5.01	7.80	5.87	9.05	5.08	8.87	4.08	6.99	4.72	7.12	4.71	7.98	4.72	7.12
11/20/2022	5.01	7.79	5.84	9.56	4.93	8.61	3.97	6.97	4.37	7.31	4.81	7.94	4.37	7.31

Bell Jet Ranger (Tonopah Block)

Date	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Detector	Summed	Summed	Summed	Summed
	110070	110070	110062	110062	10061	10061	10081	10081	110077	110077	Down	Down	Up	Up
	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137	TI208	Cs137
9/12/2022	4.99	5.50	5.27	7.02	4.59	7.78	4.56	827.00	4.68	7.51	5.04	7.69	4.68	7.51
9/13/2022	5.03	6.65	6.31	6.90	4.75	8.49	4.47	6.87	4.32	7.12	5.04	8.51	4.32	7.42
9/14/2022	5.27	7.12	6.69	8.27	4.90	8.75	4.69	8.23	4.52	4.78	5.15	8.32	4.52	4.78
9/15/2022	5.15	6.79	6.62	8.00	4.85	6.73	4.62	8.36	4.54	7.69	5.09	6.83	4.54	7.69
9/16/2022	5.17	6.79	6.70	8.41	4.77	8.54	4.55	8.23	4.55	7.68	5.00	7.13	4.55	7.68
9/17/2022	5.19	7.01	6.61	8.18	4.78	8.37	4.68	8.68	4.14	7.99	5.04	8.16	4.14	7.99
9/19/2022	5.28	8.29	6.87	8.32	4.94	6.95	4.60	6.71	4.54	7.29	5.15	6.63	4.54	7.29
9/20/2022	5.11	6.95	6.97	7.68	4.84	6.72	4.54	6.81	4.42	7.66	5.14	6.41	4.42	7.66
9/21/2022	5.17	8.32	6.84	8.20	4.92	6.93	4.48	7.11	4.32	4.99	5.12	7.08	4.32	4.99
9/22/2022	5.17	7.76	6.80	8.78	4.88	6.50	4.60	8.37	4.48	8.14	5.13	7.54	4.48	8.14
9/23/2022	5.34	7.77	6.92	8.89	4.85	8.83	4.54	7.59	4.61	6.12	5.16	8.79	4.61	6.12
9/26/2022	5.19	8.10	6.73	8.75	4.71	8.98	4.62	7.71	4.45	6.26	5.08	8.75	4.45	6.26
9/27/2022	5.17	7.37	6.66	8.80	4.75	8.81	4.56	8.41	4.59	8.06	5.09	8.63	4.59	8.06
9/28/2022	5.13	7.39	6.90	9.04	4.75	6.99	4.58	7.44	4.34	5.80	5.06	8.70	4.34	5.80
9/29/2022	5.27	6.67	6.71	8.29	4.75	5.97	4.56	8.20	4.55	7.93	5.08	8.40	4.55	7.93
9/30/2022	5.17	6.89	6.04	8.74	4.79	8.45	4.66	6.85	4.37	5.93	5.12	8.43	4.37	5.93
10/1/2022	5.16	7.35	6.74	8.44	4.79	8.77	4.63	7.02	4.24	6.09	5.11	7.63	4.24	6.09
10/2/2022	5.15	8.53	6.59	9.06	4.75	8.82	4.64	7.93	4.54	7.08	5.08	8.02	4.54	7.08
10/3/2022	5.07	7.80	6.73	8.77	4.79	7.99	4.54	7.49	4.39	6.80	5.06	8.38	4.39	6.80
10/4/2022	5.07	8.59	6.76	9.36	4.82	8.80	4.55	7.83	4.27	7.15	5.06	8.17	4.27	7.15
10/5/2022	5.35	8.66	6.82	9.16	4.83	8.77	4.63	7.57	4.27	7.04	5.19	8.70	4.27	7.04
10/6/2022	5.23	8.88	6.72	9.66	4.81	8.83	4.49	8.00	4.33	7.32	5.14	8.25	4.33	7.32